TBI Rehabilitation Through Game-Based Virtual Environment

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Abstract

Traumatic Brain Injury (TBI) results from external forces that can severely impair patient's physical and cognitive well-being. With around 70,000 annual TBI-related fatalities in the United States alone, the need for effective rehabilitation methods is pressing. However, the rehabilitation landscape presents several challenges, including the diverse impairments TBI patients face and the therapy costs. This study explores game-based interventions which offer standardized training. Our VR game, incorporating LEAP Motion, guides TBI patients to control ball in maze puzzles and collect coins, addressing physical and cognitive rehabilitation simultaneously. The game introduces features like timers, distance thresholds, reset buttons, speed sliders, and text objects for user feedback, along with a data collection toolbox to monitor progress.

Introduction

Traumatic Brain Injury (TBI) has been defined as an injury to the brain resulting from an external mechanical force, which may lead to significant impairment in the individual's physical, cognitive, and psychosocial functioning. According to statistics around 190 American, including children and adults, die from TBI-related injuries each day, totaling around 70,000 annual fatalities in the United States alone. (Center of Disease Control [1]).

Managing TBI involves various challenges such as decreasing rehabilitation effectiveness and the diverse impairments that patients suffer from. Tailoring individualized rehabilitation tasks for each patient is expensive. Studies revealed that majority of patients (75%) with traumatic brain injuries are under 35 and inclined towards playing computer and handheld games, which provides an opportunity to use game-based interventions to enhance rehabilitation efficacy and provide better treatment for TBI patients. [3].

Robotic interventions offer a comprehensive approach to therapy, promoting repetitive training and standardizing clinical practices. These devices can mimic the coordination required for daily activities, thus aiding in functional recovery.

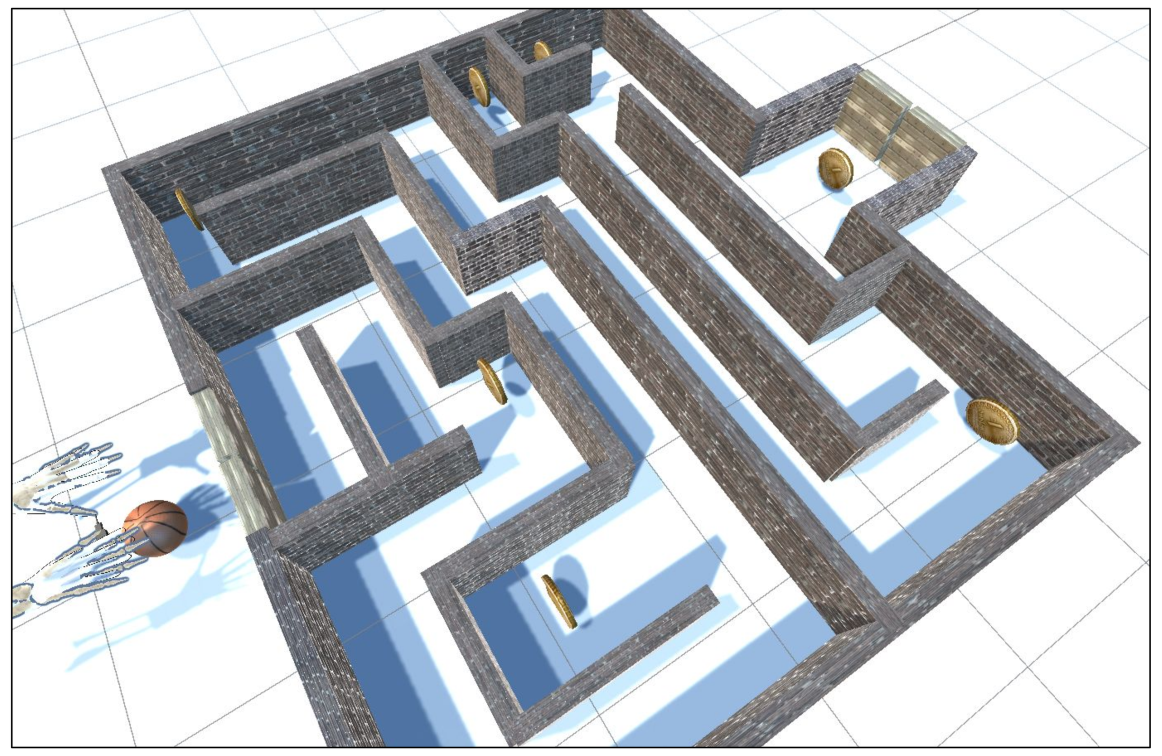

Figure 1. Scene with a panoramic view of the entire gaming environment.

This approach capitalizes on feedback mechanisms, particularly through haptic devices that simulate the manipulation of virtual objects, and leverages Virtual Reality systems, often equipped with head-mounted displays (HMDs) for a more immersive and impactful therapeutic experience.

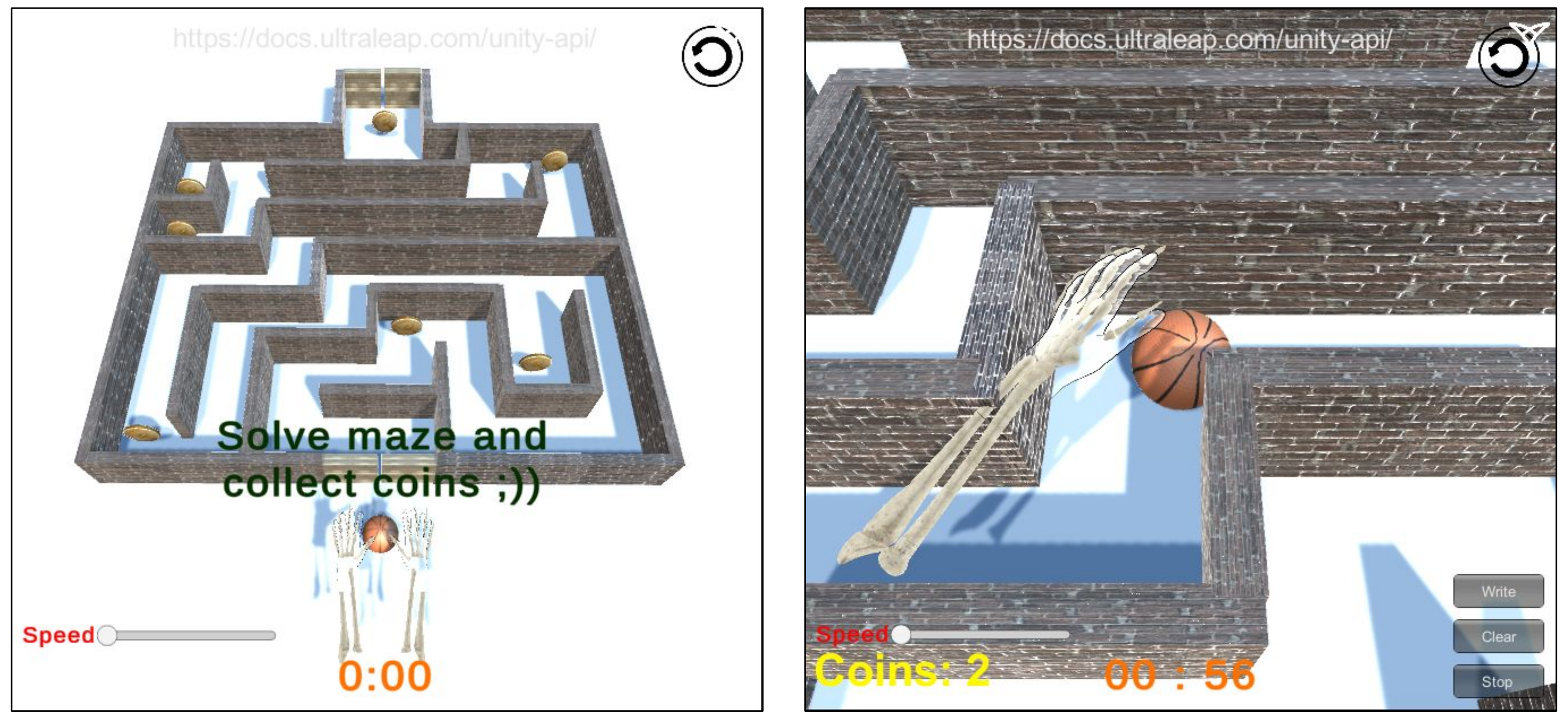

Figure 2. Game view with the task at the start (left) and with recorded coins and time remaining (right)

Methods

Our virtual game environment serves the purpose of guiding players, particularly TBI patients, through a maze puzzle while collecting coins along the way. To enhance the interactive experience, we've integrated LEAP Motion sensing technology, enabling players to control a basketball within the Unity3D game scene. This design caters to the needs of TBI patients by providing a game that simultaneously addresses their physical and cognitive rehabilitation.

The key functions implemented within the game scene include:

1. **Timer**: A timer is initiated when the basketball touches the start doors and halts when it reaches the end doors. This feature adds a time-based challenge to the game, encouraging players to navigate the maze efficiently.
2. **Distance Thresholds**: To ensure that the gameplay remains within manageable limits, XZ distance thresholds are introduced. If the basketball exceeds these predefined boundaries, the game scene is reset, and both the ball and timer are reset accordingly.
3. **Reset Button:** A reset button is provided for players to restart the game and timer if needed, offering a convenient way to retry or begin anew.
4. **Speed Slider**: Players can adjust the speed of the basketball using a slider. This feature allows for personalized gameplay experiences, catering to varying skill levels and preferences.
5. **Text Objects**: Relevant information, such as the number of collected coins and titles for the initial and ending scenes, is displayed through text objects. These elements provide players with essential feedback and context during the game.
6. **Data Collection Toolbox**: This toolbox likely includes mechanisms for gathering gameplay data, enabling the assessment of a player's performance, progression, and engagement. This data can be valuable for both players and healthcare providers in tracking rehabilitation progress.

Results

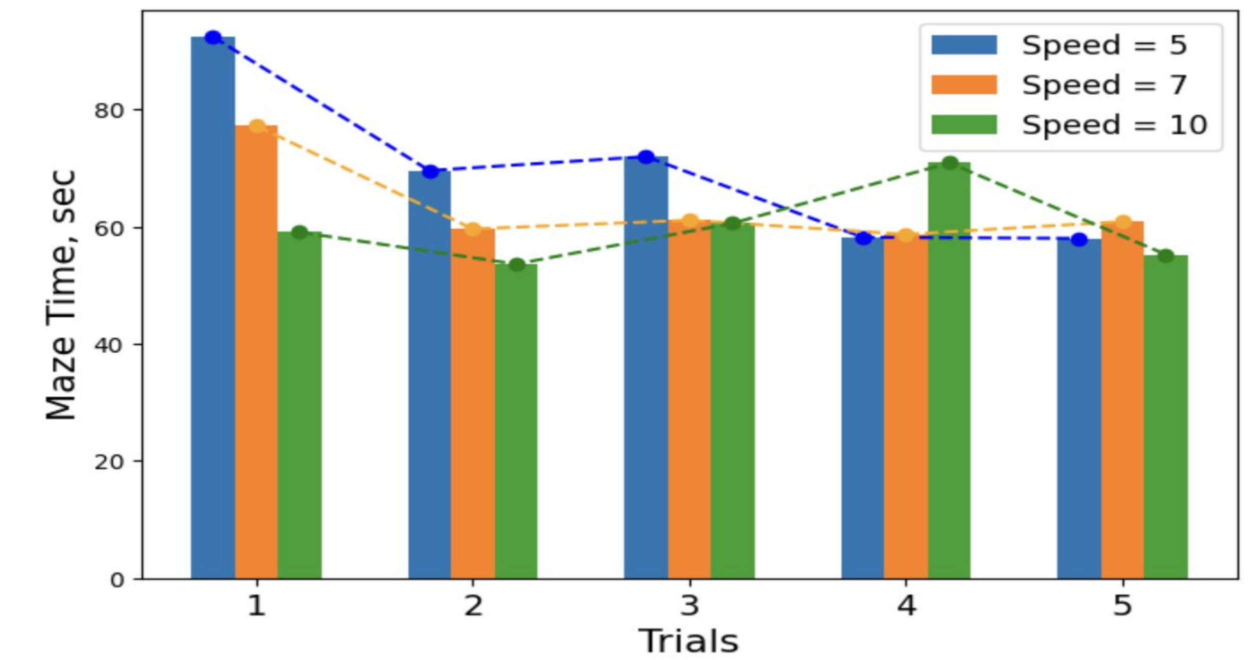

Figure 3. Maze solving time for 5 trials at different speeds Trials 4,5 for speed=5 for 4 coins, while other trials with 6 coins

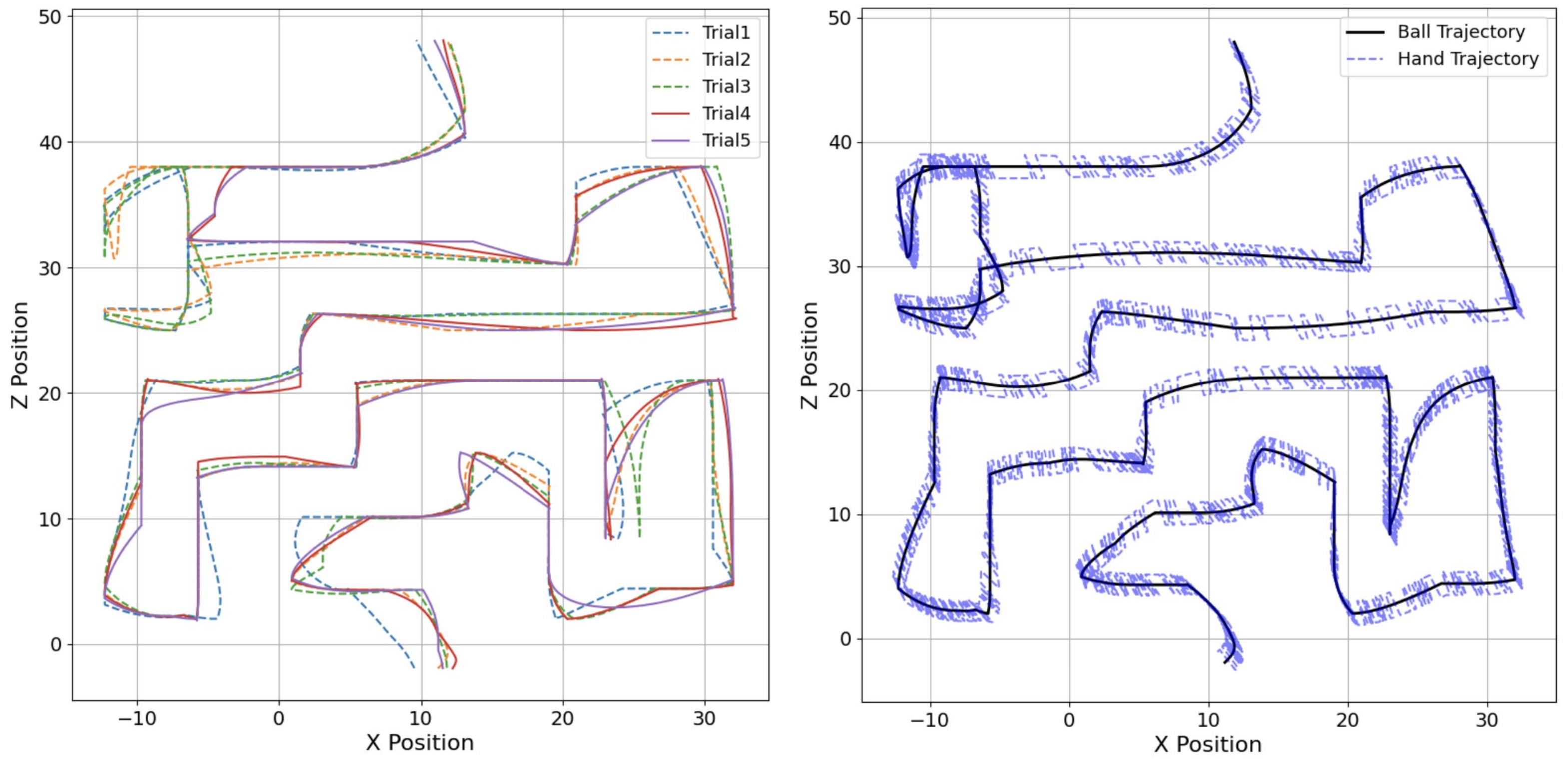

Figure 4. Ball trajectory in maze for each trial (left) and mean LEAP motion hand and ball trajectories (right)

Conclusion

The future development of app could be enhanced by the introduction of additional maze levels with varying complexities and the option to change the ball's mass and dimensions (tennis, soccer balls). In terms of hardware integration, incorporating IMUs to measure hand speed and visually represent it on the grid map can enhance user engagement. Vibro motors and LEDs can provide visual and tactile feedback to patients when they collect coins. These features not only add depth to the virtual environment but also provide more comprehensive and personalized rehabilitation experiences for TBI patients.

References

1. TBI Data | Concussion | Traumatic Brain Injury | CDC Injury Center
2. Shen, Jiabin, et al. "Virtual reality–based executive function rehabilitation system for children with traumatic brain injury: design and usability study." JMIR Serious Games 8.3 (2020): e16947.
3. Shapi'i, Azrulhizam, Nor Azan Mat Zin, and Ahmed Mohammed Elaklouk. "A game system for cognitive rehabilitation." BioMed research international 2015 (2015).
4. Jacoby, Michele, et al. "Effectiveness of executive functions training within a virtual supermarket for adults with traumatic brain injury: a pilot study." IEEE transactions on neural systems and rehabilitation engineering 21.2 (2012): 182-190.